End-sem: worked solutions

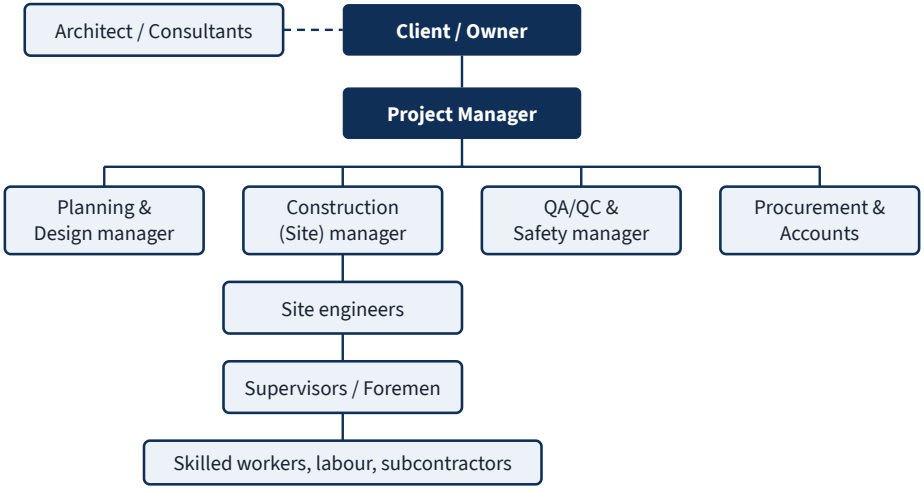
13 Nov 2025 · 40 marks · 3 hours · CEPC11

The paper is printed "13.11.2024", but it sits in the Nov 2025 end-semester timetable and came with the 2025 Section B papers, so it is filed under 2025.

Q1

1 mark

> Draw the organizational chart of a construction project.



The **client** appoints the consultants and the **project manager**, who leads the planning/design, site, QA/QC and safety, and procurement/accounts functions. Authority flows down from site manager → site engineers → supervisors → workers.

Q2

1 mark

> The addition of pozzolana to cement causes: a) less heat of hydration b) more heat of hydration c) increase in strength d) decrease in workability

(a) Less heat of hydration. The pozzolan replaces part of the cement and reacts slowly, so heat is released slowly. (Strength increases only at later ages; early strength is lower.)

Q3

1 mark

> For a given environment, the most significant factor that influences the total shrinkage of concrete is: a) cement content b) total amount of water at the time of concreting c) aggregate size d) proportioning of fine aggregate

(b) Total amount of water at the time of concreting. Shrinkage comes from the loss of water from the paste, so the **water content per unit volume** controls it most strongly (for a given environment and aggregate).

Q4

2 marks

> Write the different feasibility studies carried out during project selection.

- **Technical feasibility**: site conditions, soil, technology, materials, design viability.
- **Economic / financial feasibility**: cost–benefit, NPV, IRR, payback, funding sources.
- **Market (demand) feasibility**: need for the facility, expected demand and revenue.
- **Legal and regulatory feasibility**: land title, approvals, zoning, permits, codes.
- **Environmental feasibility**: environmental impact assessment, clearances.
- **Social feasibility**: impact on community, resettlement, public acceptance.
- **Schedule (time) and operational feasibility**: can it be built and run within the required time and resources.

Q5

2 marks

> Describe Bogue's components and their percentage in ordinary Portland cement.

Compound	Name	Typical %	Role
C_3S	Tricalcium silicate (alite)	45–60	Early strength, high heat
C_2S	Dicalcium silicate (belite)	15–30	Later strength, low heat
C_3A	Tricalcium aluminate	6–12	Flash set (controlled by gypsum), highest heat, poor sulfate resistance
C_4AF	Tetracalcium aluminoferrite	6–8	Little strength, colour

Bogue equations (oxide %): $C_3S = 4.071C - 7.600S - 6.718A - 1.430F - 2.852SO_3$; $C_2S = 2.867S - 0.7544\ C_3S$; $C_3A = 2.650A - 1.692F$; $C_4AF = 3.043F$.

Q6

2 marks

> What are all the tests conducted for the quality of cement in the laboratory?

Physical tests (IS 4031): fineness (Blaine air permeability / sieve), **standard consistency**, **initial and final setting time** (Vicat), **soundness** (Le Chatelier, autoclave), **compressive strength** (70.6 mm mortar cubes at 3, 7, 28 days), heat of hydration, specific gravity, drying shrinkage.

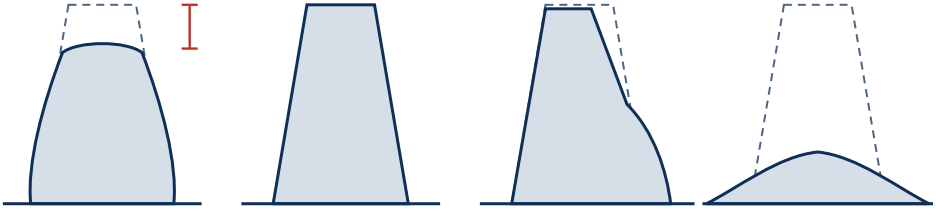
Chemical tests (IS 4032): loss on ignition, insoluble residue, MgO, SO_3, chlorides, total alkalis, lime saturation factor, alumina/iron ratio.

Q7

2 marks

> Define workability of concrete. What are the types of slump, with picture? Compare the compaction factor test and the slump cone test.

Workability: the ease with which fresh concrete can be mixed, placed, compacted and finished without segregation.



True slump
subsides evenly
valid result

Zero slump
keeps cone shape
very dry mix

Shear slump
one side slides off
repeat the test

Collapse slump
spreads out
very wet mix

	Slump cone test	Compacting factor test
Measures	Subsidence of a cone of concrete under its own weight	Degree of compaction from a standard fall (partial ÷ full compacted mass)
Best for	Medium–high workability (25–125 mm)	**Low workability** (CF 0.7–0.95)
Sensitivity	Insensitive for stiff mixes; shear/collapse possible	**More precise and sensitive**
Where	**Site** (simple, quick, cheap)	Mainly **laboratory**
Limits	Not for aggregate > 40 mm or very dry/wet mixes	Bulky apparatus; not for very wet mixes

Q8

2 marks

> What are all the main compounds to be present in a material to be added as a mineral admixture for cement replacement? Why?

The material must contain **reactive (amorphous / glassy) silica, SiO_2**, and usually **alumina, Al_2O_3** (with some Fe_2O_3); for GGBS and Class C fly ash also **lime (CaO)**.

- Typical requirement (ASTM C618): **$SiO_2 + Al_2O_3 + Fe_2O_3 \geq 70\%$** (Class F fly ash) or ≥ 50% (Class C), with low LOI (≤ 6%) and SO_3 (≤ 5%).

Why: reactive SiO_2 and Al_2O_3 react with the **$Ca(OH)_2$** released by cement hydration (pozzolanic reaction) to form **additional C-S-H and C-A-H**, the strength-

giving products. Crystalline silica (e.g. quartz sand) does not react, so the silica must be **amorphous and finely divided**.

Q9 (a)

2 marks

> Design M15 concrete via nominal mix for fine aggregate of zone II and coarse aggregate of 20 mm size.

IS 456 Table 9 for **M15**: **350 kg total dry aggregate per 50 kg cement**, FA : CA = **1 : 2** (zone II sand, 20 mm CA), water **32 litres per 50 kg cement**.

- Per 50 kg cement: FA = **116.7 kg**, CA = **233.3 kg**, water = **32 l** (w/c 0.64).
- Ratio ≈ **1 : 2 : 4** (by mass, more exactly 1 : 2.33 : 4.67).
- Per m^3 (approx., 2% air): cement **275 kg (5.5 bags)**, FA **641 kg**, CA **1282 kg**, water **176 l**.

Q9 (b)

2 marks

> What is meant by compaction? Explain the various methods of compaction. Explain the effects of compaction.

Compaction removes entrapped air from fresh concrete and packs the particles together to achieve **maximum density**.

Methods:

- **Manual**: rodding (16 mm rod, layers 15–20 cm), ramming (plain concrete at ground level), tamping (wooden beam, slabs and roads).
- **Mechanical (vibration)**: internal/needle (poker) vibrators, external/form vibrators, vibrating tables, surface/screed vibrators; also centrifugation, pressure and jolting for precast work.

Effects:

- Proper compaction → high density, **higher strength** (each 1% of air left lowers strength by about 5–6%), lower permeability, better bond with steel and better durability; no honeycombing.

- **Under-vibration**: honeycombing, entrapped air voids, sand streaks, placement lines, cold joints.
- **Over-vibration**: segregation, bleeding, laitance at the top, formwork damage.

Q10

2 marks

> List the different types of placing of concrete and the precautions to take during placing.

Methods of placing: by hand (pans, buckets), wheelbarrows/buggies, chutes, skips and hoists, belt conveyors, **pumping**, cranes and buckets, **tremie (under water)**, **slip-forming**, **shotcrete (dry/wet mix)**, preplaced aggregate grouting.

Precautions:

- Deposit concrete **as near as possible to its final position**; do not move it sideways with vibrators.
- **Limit free fall** (about 1.5 m); use chutes or drop pipes to avoid segregation.
- Place in **uniform horizontal layers** (300–450 mm) and compact each layer before the next; place the next layer **while the lower one is still plastic** (no cold joints).
- **Rate of placing = rate of compaction**.
- Formwork clean, oiled, tight and wetted; reinforcement fixed with cover blocks.
- Place within the initial setting time; avoid placing in heavy rain or extreme heat without precautions.

Q11

2 marks

> Explain briefly the various non-destructive tests for concrete.

- **Rebound hammer (IS 13311 Part 2)**: a spring-loaded hammer strikes the surface; the rebound number relates to surface hardness and, via a calibration curve, to compressive strength.

- **Ultrasonic pulse velocity (IS 13311 Part 1)**: the travel time of an ultrasonic pulse through the concrete gives the velocity (> 4.5 km/s excellent, 3.5–4.5 good, 3.0–3.5 medium, < 3.0 doubtful); detects cracks, voids and honeycombing.
- **Penetration resistance (Windsor probe)** and **pull-out test**: force needed to drive in or pull out a probe/insert.
- **Cover meter / rebar locator**: cover and bar position.
- **Half-cell potential**: probability of reinforcement corrosion.
- **Radiography and ground-penetrating radar**: internal defects and bar layout.
- **Core test** (partially destructive): direct strength of in-place concrete.

Q12

3 marks

> Explain in detail any 3 types of special concrete, with the change in mix design with respect to ordinary concrete mix design.

Special concrete	What it is	Change in mix design compared with ordinary concrete
Self-compacting concrete (SCC)	Flows under its own weight, fills congested forms without vibration, no segregation	**Higher powder content** (450–600 kg/m^3 incl. fly ash/limestone powder); **SP + VMA**; **less coarse aggregate** (about 50% of solid volume) and smaller max. size (10–20 mm); more fine aggregate. Designed by **slump flow, V-funnel, L-box**, not slump
High-strength concrete (HSC)	$f_{ck} \geq 60$ MPa	**Very low w/b (0.25–0.35)**; high binder content; **silica fume** 5–15%; high dose of **superplasticizer**; **smaller, strong aggregate (10–12.5 mm)**; careful curing
Fibre-reinforced concrete (FRC)	Steel, glass, polypropylene fibres (0.5–2% by volume) give toughness and crack control	**Higher paste and fines content**, smaller max. aggregate, more SP to restore workability lost to fibres; workability checked by Vebe/inverted slump cone
Lightweight concrete	Density 300–1850 kg/m^3	Lightweight aggregate (pre-wetted), design by volume, more cement

Special concrete	What it is	Change in mix design compared with ordinary concrete
Mass concrete	Dams, thick rafts	Low-heat cement / high fly ash or slag, low cement content, large aggregate, pre-cooling

Q13

3 marks

> Mix designed for M35: cement 400 kg, fine aggregate 728 kg, coarse aggregate 1186 kg, water 167 litres. Water absorption of FA and CA: 1% and 0.5%. After heavy rain the moisture content of FA and CA is 2% and 5%. Which weights must be corrected, and by how much?

The design quantities are for **SSD** aggregate. Free (surface) moisture = total moisture − absorption.

	FA	CA
Free moisture	2 − 1 = 1%	5 − 0.5 = 4.5%
Free water brought in	0.01 × 728 = **7.28 kg**	0.045 × 1186 = **53.37 kg**
Corrected (wet) mass to weigh	728 + 7.28 = **735.3 kg**	1186 + 53.37 = **1239.4 kg**

Corrected water = 167 − 7.28 − 53.37 = **106.4 litres**. Cement stays **400 kg**.

Corrections: FA **+7.3 kg**, CA **+53.4 kg**, water **−60.6 litres**. Without the correction the effective w/c would rise from 0.42 to about 0.57 and strength would fall sharply.

Q14

3 marks

> Mention the steps that can be taken (with reasoning) for: (i) reducing alkali–aggregate reaction, (ii) concreting in regions of high temperature, and (iii) reducing permeability of concrete without affecting workability in the fresh stage.

(i) Reducing alkali–aggregate reaction

- Use **non-reactive aggregate** (test by petrography / mortar bar).

- Use **low-alkali cement** (Na_2O eq. < 0.6%) and limit total alkali in concrete (< 3 kg/m^3).
- Replace cement by **fly ash, GGBS, silica fume or metakaolin**: they dilute alkalis and bind them in low C/S C-S-H.
- **Lithium compounds** (LiOH, LiNO3) form non-expansive gel.
- **Reduce moisture access** (dense concrete, coatings, drainage), since the gel swells only with water.

(ii) Hot-weather concreting

- **Cool the ingredients**: chilled water or ice, shaded/sprinkled aggregate, cool cement; aim for concrete below about 32 °C.
- Use **retarders** (or retarding SP) to delay setting and prevent cold joints; use low-heat or blended cement.
- **Place quickly**, at night or in the cooler part of the day; wet the forms and subgrade.
- **Prevent evaporation** (windbreaks, fog sprays, curing compound) and **start curing immediately** to avoid plastic shrinkage cracks.

(iii) Reducing permeability without losing workability

- **Lower the w/c using a superplasticizer**: less water → fewer capillary pores, while the SP keeps the slump.
- **Mineral admixtures** (fly ash, GGBS, silica fume, metakaolin): pore refinement and denser ITZ; fly ash also improves workability.
- **Good grading** (dense packing), **proper compaction** and **adequate moist curing**.
- Optionally integral **waterproofing admixtures** (hydrophobic) that do not affect fresh properties.

Q15

3 marks

> A roof slab containing 10 m^3 of M40 concrete is laid on the new civil annex building. As a quality inspector, how many cube samples will you prepare? Cubes are tested after 7 and 28 days. Based on the results, comment on the strength and acceptability as per IS 456.

Specimen	7-day load (kN)	28-day load (kN)
CA 25/9/23 SP_1	605	821
CA 25/9/23 SP_2	755	952
CA 25/9/23 SP_3	735	1015

Number of samples (IS 456 Cl. 15.2.2): 6–15 m^3 → **2 samples**; each sample = 3 cubes, with extra cubes for 7-day testing. So **2 × 3 = 6 cubes for 28 days + 6 cubes for 7 days = 12 cubes** (at least one sample per shift).

Strengths (150 mm cubes, area 22,500 mm^2):

Specimen	7-day (MPa)	28-day (MPa)
SP_1	26.89	36.49
SP_2	33.56	42.31
SP_3	32.67	45.11
Mean (test result)	**31.04**	**41.30**

- **Validity**: deviations from the mean are −13.4/+8.1/+5.3% (7 d) and −11.7/+2.4/+9.2% (28 d), all within **±15%** → valid.
- **7-day**: 31.04 / 41.30 = 75% of the 28-day value, and above 0.67 × 40 = 26.8 MPa → normal gain.
- **Individual criterion**: 41.30 ≥ f_{ck} − 4 = **36 MPa** ✓.
- **Mean criterion**: needs ≥ f_{ck} + 4 = 44 MPa (or 40 + 0.825 × 5 = 44.1) for a group of 4 results. 41.30 is **below 44**, and well below the target mean of 48.25 MPa.

Comment: the concrete exceeds its characteristic strength and passes the individual criterion, so it is **acceptable**, but only marginally. The result is below

the target mean strength, so the mix/quality control should be reviewed, and further results must be tracked to check the mean criterion.

Q16

3 marks

> A simply supported beam of length 5 m and cross-section 300 × 500 mm is cast using M40 concrete. The 7-day cube results are as in Q15. Can the formwork be removed fully within 7 days? Justify with calculation.

Stress in the beam from self-weight when the forms are removed:

- Self-weight w = 0.3 × 0.5 × 25 = **3.75 kN/m**
- M = $wL^2/8$ = 3.75 × 5^2 / 8 = **11.72 kN·m**
- Z = $bd^2/6$ = 300 × 500^2 / 6 = **12.5 × 10^6 mm^3**
- Bending stress f = M/Z = 11.72 × 10^6 / 12.5 × 10^6 = **0.94 MPa**

Strength available at 7 days: f_c = **31.04 MPa** (78% of f_{ck}); flexural tensile strength ≈ 0.7 √31.04 = **3.9 MPa**.

IS 456 Cl. 11.3 allows formwork to be removed when the concrete has at least **twice the strength needed to carry the stresses** at that time: 2 × 0.94 = 1.88 MPa, which is far less than both 31 MPa (compression) and 3.9 MPa (tension). The common site rule of 70% of f_{ck} is also met (31.04 > 28 MPa).

Conclusion: by calculation, **yes** for self-weight alone: the side forms and soffit forms can be removed at 7 days. However, IS 456 Cl. 11.3.1 recommends that **props under beams spanning up to 6 m stay for 14 days** (soffit forms at 7 days only with props re-fixed). So remove the side and soffit forms at 7 days, but **re-prop** the beam if any construction or slab load is to be placed on it before 14 days.

Q17

4 marks

> Design an M50 concrete mix for a bridge column member. Use OPC 53 grade cement. Specific gravity of fine aggregate, coarse aggregate and cement are 2.65, 2.8 and 3.1. Water absorption of coarse and fine aggregate is 1% and 1.5%. Use coarse aggregate of 10 mm crushed sub-angular aggregate. Choose workability from IS 456. Specific gravity of superplasticizer is 1.15; of fly ash and silica fume 2.2 and 2.3.

Assumptions: severe exposure (bridge column in the open): max w/c 0.45, min cement 320 + 40 (10 mm aggregate) = 360 kg/m^3. Heavily reinforced column → **medium workability, slump 100 mm** (IS 456 Cl. 7.1: 50–100 mm). Binder: **20% fly ash + 8% silica fume**. SP 1% of binder, 20% water reduction. Zone II sand. Aggregates dry (absorption correction applied). Good supervision.

Step	Working	Result
Target strength	max(50 + 1.65 × 5, 50 + 6.5)	**58.25 MPa**
w/b	Fig. 1, IS 10262:2019 (OPC 53) → 0.32 (< 0.45 ✓)	**0.32**
Water	10 mm: 208 kg; sub-angular −10 = 198; +6% (100 mm) = 209.9; −20% (SP)	**167.9 kg/m^3**
Binder	167.9 / 0.32	**524.7 kg/m^3**
Split	OPC 72% / fly ash 20% / silica fume 8%	**377.8 / 104.9 / 42.0 kg** (OPC < 450 ✓)
SP	1% × 524.7	**5.25 kg**
CA fraction	10 mm, zone II: 0.46 + (0.5 − 0.32)/0.05 × 0.01	**0.496**

Volume (m^3)	
Air	0.0100
OPC 377.8 / 3100	0.1219
Fly ash 104.9 / 2200	0.0477
Silica fume 42.0 / 2300	0.0183

Volume (m^3)	
Water	0.1679
SP 5.25 / 1150	0.0046
All-in aggregate	**0.6297**
CA = 0.6297 × 0.496 × 2.8 × 1000	**875 kg** (SSD)
FA = 0.6297 × 0.504 × 2.65 × 1000	**841 kg** (SSD)

Absorption correction (dry aggregates): dry CA = 875/1.01 = **866 kg**; dry FA = 841/1.015 = **829 kg**; extra water = 21.1 kg → **water to add = 189 kg**.

Final mix per m^3: OPC **378 kg**, fly ash **105 kg**, silica fume **42 kg**, water **189 kg** (free w/b 0.32), FA **829 kg**, CA 10 mm **866 kg**, SP **5.3 kg**. Ratio (binder : FA : CA, SSD) = **1 : 1.60 : 1.67**. Verify by trial mixes.